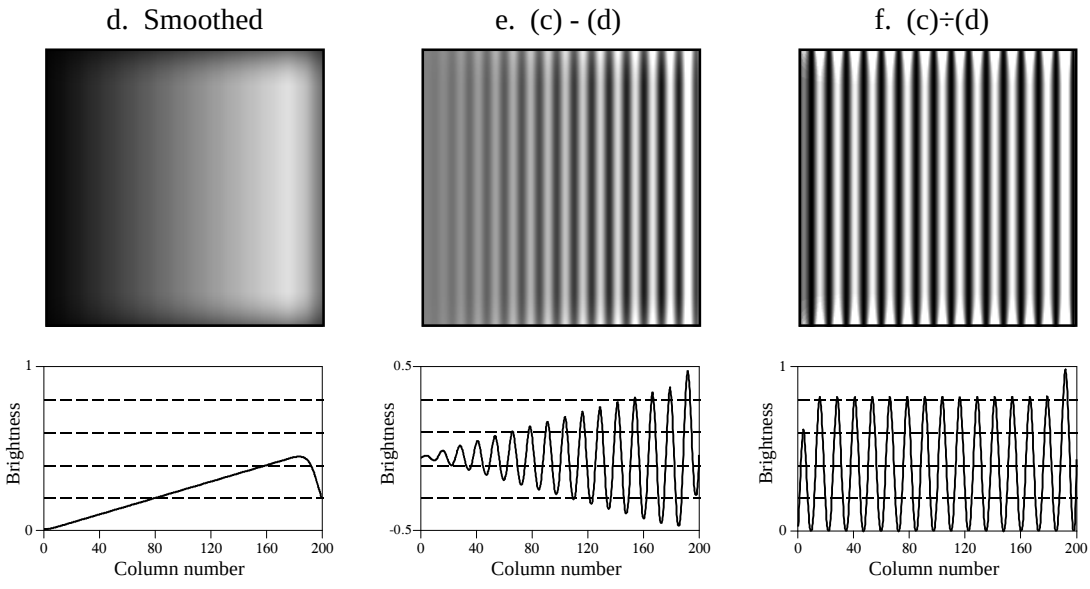


FIGURE 24-8 (continued)
Figure (d) is a smoothed version of (c), used as an approximation to the illumination signal. Figure (e) shows an approximation to the reflectance image, created by *subtracting* the smoothed image from the viewed image. A better approximation is shown in (f), obtained by the nonlinear process of *dividing* the two images.

multiplication, homomorphic processing starts by taking the *logarithm* of the acquired signal. With the identity: $\log(a \times b) = \log(a) + \log(b)$, the problem of separating *multiplied* signals is converted into the problem of separating *added* signals. For example, after taking the logarithm of the image in (c), a linear high-pass filter can be used to isolate the logarithm of the reflectance image. As before, the quickest way to carry out the high-pass filter is to subtract a smoothed version of the image. The antilogarithm (exponent) is then used to undo the logarithm, resulting in the desired approximation to the reflectance image.

Which is better, dividing or going along the homomorphic path? They are nearly the same, since taking the logarithm and subtracting is *equal* to dividing. The only difference is the approximation used for the illumination image. One method uses a smoothed version of the acquired image, while the other uses a smoothed version of the logarithm of the acquired image.

This technique of flattening the illumination signal is so useful it has been incorporated into the neural structure of the eye. The processing in the middle layer of the retina was previously described as an edge enhancement or high-pass filter. While this is true, it doesn't tell the whole story. The first layer of the eye is nonlinear, approximately taking the *logarithm* of the incoming image. This makes the eye a homomorphic processor. Just as described above, the logarithm followed by a linear edge enhancement filter flattens the illumination component, allowing the eye to see under poor lighting conditions. Another interesting use of homomorphic processing